



Started on	Thursday, 20 August 2026, 8:48 AM
State	Finished
Completed on	Thursday, 20 August 2026, 8:54 AM
Time taken	5 mins 48 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void Factor(int num) {
4      int count = 0;
5
6      count++;
7      for (int i = 1; i <= num; ++i) {
8          count++;
9          if (num % i == 0) {
10             count++;
11         }
12         count++;
13     }
14
15     printf("%d", count);
16 }
17
18 int main() {
19     int n;
20     if (scanf("%d", &n) == 1 && n > 0) {
21         Factor(n);
22     }
23     return 0;
24 }
25
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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